

A be a set.

A relation on set A is a subset of the $\mathcal{P}(A \times A)$.

Ex: $A = \{1, 2, 3\}$

$$A \times A = \{ \underline{(1,1)}, \underline{(1,2)}, \underline{(1,3)}, \underline{(2,1)}, \underline{(2,2)}, \underline{(2,3)}, \underline{(3,1)}, \underline{(3,2)}, \underline{(3,3)} \}$$

$$P_A = \{ \underline{(1,1)}, \underline{(1,2)} \} \times \begin{matrix} 2x \\ 3x \end{matrix}$$

$$\mathcal{P}(\{a, b, c\}) = \{ \varnothing, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{c, a\}, \{a, b, c\} \}$$

$P_A \subseteq A \times A$

$$\underline{P_A} = \{ \underline{(1,1)}, \underline{(2,1)}, \underline{(3,2)} \} \subseteq A \times A$$

$$\underline{1P_A1}, 2P_A1, 3P_A2$$

$$P_A : A \rightarrow A$$

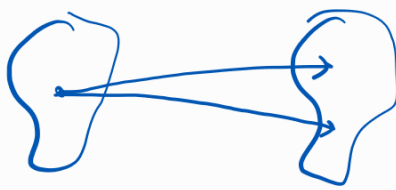
$$\begin{matrix} P_A(1) = 1 \\ P_A(2) = 1 \\ P_A(3) = 2 \end{matrix}$$

$$f_A \subseteq A \times A, f_A : A \rightarrow A$$

- ① Domain all element should be mapped.
- ② one element must be mapped to a unique element.

$$f_A(1) = 1$$

$$f_A(1) = 2 \times$$



$$P_{A,B} \subseteq A \times B.$$

Q1 $S = \mathbb{Z} \times \mathbb{Z}$

$$P_S = S \times S = (\mathbb{Z} \times \mathbb{Z}) \times (\mathbb{Z} \times \mathbb{Z})$$

$$P_S := (\tilde{a}, \tilde{b}) P_S (\tilde{c}, \tilde{d}) \quad ((\tilde{a}, \tilde{b}), (\tilde{c}, \tilde{d})) \in \underline{P_S} \subseteq S \times S$$

$\uparrow \quad \quad \uparrow$
 $\mathbb{Z} \times \mathbb{Z} \quad \mathbb{Z} \times \mathbb{Z}$
 $\parallel \quad \parallel$
 $S \quad S$

$a + d = b + c$

① Reflexive: $(\tilde{a}, \tilde{b}) P_S (\tilde{a}, \tilde{b})$ S

?

$$a + b = b + a$$

$\uparrow \quad \uparrow \quad ? \quad \uparrow \quad \uparrow$
 $\mathbb{Z} \quad \mathbb{Z} \quad ? \quad \mathbb{Z} \quad \mathbb{Z}$
 $\underline{\quad} \quad \underline{\quad} \quad \quad \underline{\quad} \quad \underline{\quad}$

$$M_2 \times M_2 = S$$

$$(A, B) P_S (C, D)$$

$$AB = BA$$

$$\Leftrightarrow AD = BC$$

② $(\tilde{a}, \tilde{b}) P_S (\tilde{c}, \tilde{d}) \stackrel{?}{\Rightarrow} (\tilde{c}, \tilde{d}) P_S (\tilde{a}, \tilde{b})$

$$\Rightarrow \underline{a+d} \leq \underline{b+c} \Leftrightarrow \underline{c+b} = \underline{d+a}$$

$$(3) \quad (\underline{a}, \underline{b}) \rho_s (\underline{c}, \underline{d}) \wedge (\underline{c}, \underline{d}) \rho_s (\underline{e}, \underline{f})$$

$$\stackrel{?}{\Rightarrow} (\underline{a}, \underline{b}) \rho_s (\underline{e}, \underline{f})$$

$$\checkmark$$

$$c+f = d+e$$

$$\Rightarrow f = d+e-c$$

$$a+d = b+c$$

$$\Rightarrow a = b+c-d$$

$$(a+f = b+e)$$

$$a+f = b+\cancel{c-\cancel{c}} + \cancel{d} + e - \cancel{d}$$

$$= b+e.$$

$$\checkmark \quad \underline{A \cup B} = \underline{(A \cap B) \Delta (A \Delta B)} \quad \left[A \Delta B = (A \cup B) - \underline{(A \cap B)} \right]$$

$$\text{pt. } A = B$$

$$(1) \quad \leftarrow \underline{A \subseteq B}$$

$$(2) \quad \underline{B \subseteq A}$$

$$\searrow \quad \underline{A = B}$$



$$(1) \quad A \cup B \subseteq \underline{(A \cap B) \Delta (A \Delta B)}$$

$$\swarrow$$

$$\underline{x \in A \cup B}$$

$$\Rightarrow x \in A \text{ or } x \in B$$

✓

$$B \cup (A - B)$$

$$= A \quad (\because B \subseteq A)$$

↓

$$B \cup (A - B) = (A \cap B^c) \cup B$$

$$= (A \cup B) \cap (B^c \cup B)$$

$$= A \cap S$$

$$= \underline{\underline{A}}$$

$$(A \cap B) \Delta (A \Delta B)$$

$$((A \cap B) \cup (A \Delta B)) \setminus$$

$$((A \cap B) \cap (A \Delta B))$$

$\emptyset$

$$(A \cup B) \setminus (A \cap B)$$

$$(A \cap B) \cup (A \Delta B)$$

$$(A \cap B) \cup ((A \cup B) - (A \cap B))$$

$$= \underline{\underline{A \cup B}}$$

